

When Payoff Returns: Reinstatement of Earlier Narrative Context During Recall

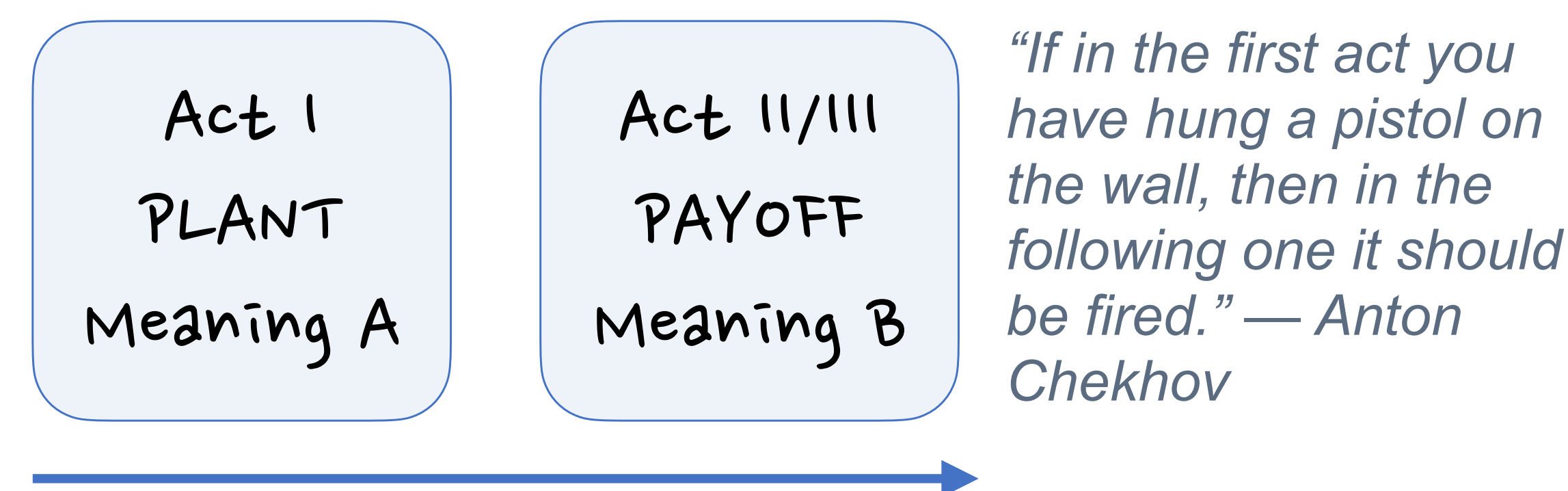
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Takeaway

- During viewing, plant-payoff pairs were not detectable as sustained pattern similarity.
- When participants recall a payoff later, posterior medial cortex (PMC) patterns resemble the pattern of its own plant more than other plants.

Background

Does the brain hold on to the link between a plant and its payoff?



Plant and payoff

- Concept from screenwriting
- Usually **don't** function as boundaries
- Usually **across** events

Research Questions

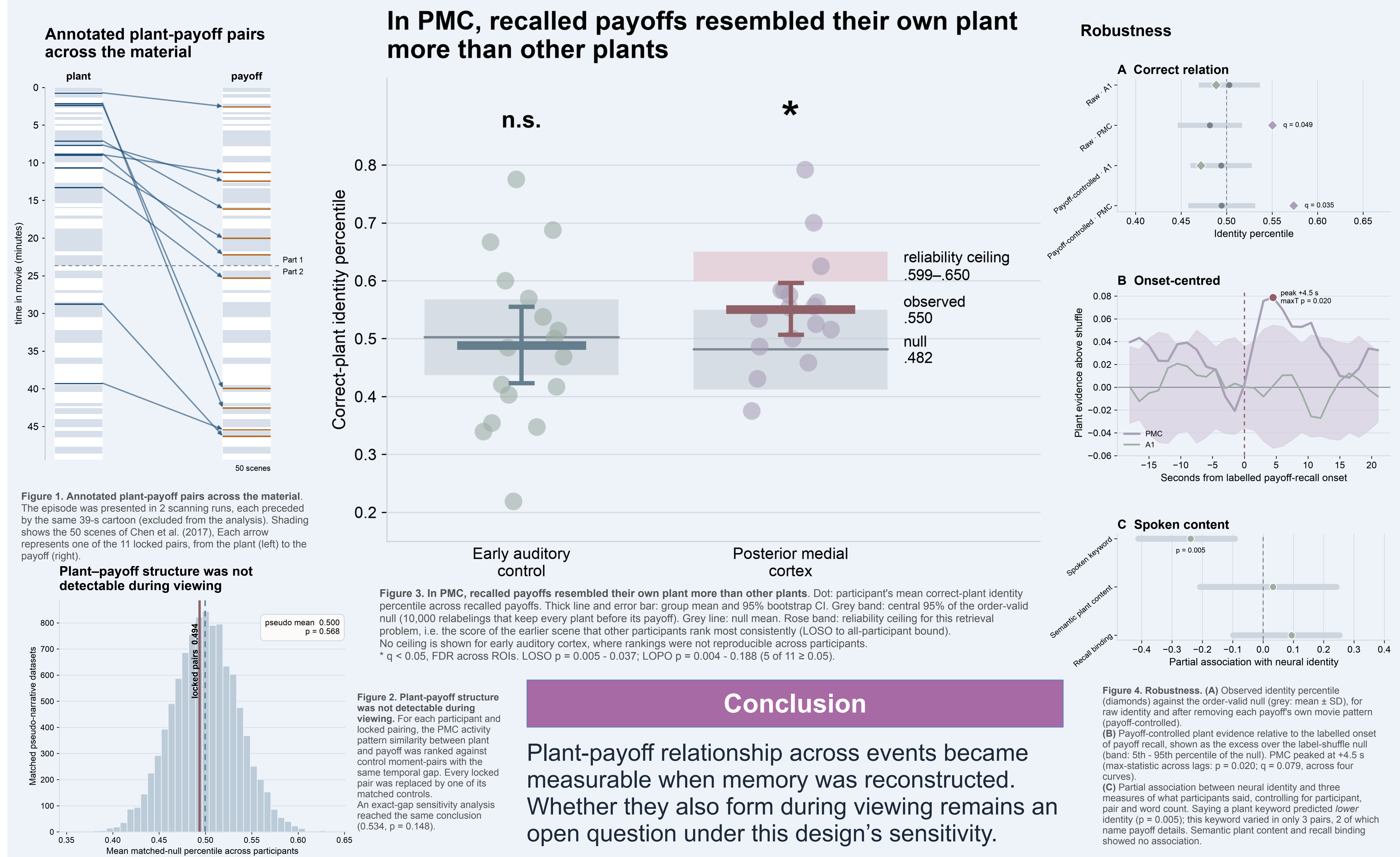
- As meaning changes, does the brain encode the plant and the payoff in **similar patterns**?
- How is the plant-payoff relationship expressed when the payoff is recalled?

Materials & Methods

- Data: Sherlock (Chen et al., 2017).
N = 17; Sherlock S1E1, **Viewing + Free Recall, fMRI**
- Locked** annotated plant-payoff pairs, **N = 11** (modality, timing, cue, characters)
One annotator



Results



Conclusion

Plant-payoff relationship across events became measurable when memory was reconstructed. Whether they also form during viewing remains an open question under this design's sensitivity.

Discussion

Viewing

No stable plant-payoff similarity within this design's sensitivity.

Recall

Recalling a payoff brings back its own plant's context in PMC, around recall onset.

Next steps

Across datasets
Number of annotators
Other storytelling techniques

References

Chen J, et al. Shared memories reveal shared structure in neural activity across individuals. *Nat Neurosci.* 2017;20:115–125.
Baldassano C, et al. Discovering event structure in continuous narrative perception and memory. *Neuron.* 2017;95:709–721.

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